



Mr. Ernest Kwabena Asare

Software Engineer | MSc. Data Analytics Student

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🌐 Ernest Kwabena Asare 🔗 quantum-ernest 🔗 Portfolio

👤 ABOUT ME

Backend-focused Software Engineer with experience in building and scaling automation, IoT, and ERP systems in production environments. Proven track record of delivering high-impact solutions as a team lead, designing real-time data systems, integrating platforms, and driving cross-functional development efforts. Currently pursuing an MSc in Data Analytics with a focus on Machine Learning.

🧠 SKILLS

Back-End: Python, Go, JavaScript, PyTorch, TensorFlow | **Front-End:** React | **Tools:** Git, AWS, Ansible, Grafana, Kafka, Docker, Prometheus, Looker Studio, Elasticsearch, Nginx, Linux, Github Actions | **Databases:** PostgreSQL, MongoDB, Redis

🎓 EDUCATION

MSc. Data Analytics 2026 – Present
Universität Hildesheim
Specialization: Machine Learning

Bsc. Computer Science 2017 – 2021
University of Ghana

👛 PROFESSIONAL EXPERIENCE

The Quantum Group Limited 11/2022 – 11/2025
Automation (Team Lead) & Systems Engineering

- **Odoo ERP:** Engineered custom Odoo modules and integration layers, building connectors to external systems (payments, IoT, APIs), contributing to cross-version migrations, and extending core functionalities to deliver scalable, business-critical ERP solutions.
- **Fuel Station Automation:** Implemented a real-time Fuel Station Management and Monitoring system that reduced average reporting time and station losses by 25% while improving operational efficiency. Developed an ERP solution for configurations, reporting, and price adjustments, accelerating decision-making processes.
- **Remcash Payment Gateway:** Integrated Telecel Cash API and developed secure encryption and compliance services, achieving full AML & KYC compliance. Created a streamlined onboarding system that reduced customer onboarding time by 40%.
- **Fuel Truck Automation System:** Engineered a real-time tracking solution for GPS, temperature, and fuel volume data in fuel trucks, improving fuel management and preventing theft. Integrated ERP applications for seamless device synchronization and efficient management.

- **Lunch Ordering System:** Designed and developed backend services for the system serving over 1,000 employees, ensuring robust performance, efficiency, and scalability.
- **Fleet Management System:** Devised and built a robust system supporting the Galileosky 7x device, with features like real-time tracking, automated notifications, driver metrics, and downloadable vehicle history. This system currently manages over 100 vehicles across multiple locations.
- **Cylinder Bottling Automation:** Led the development of a QR code reader system, increasing data accuracy and reducing production time by 45% using RS-485 serial communication.

University of Ghana Computing Systems

10/2021 – 10/2022

IT Security Specialist

- Researched social engineering attacks, identified vulnerabilities, and implemented countermeasures. Analyzed survey data to improve student experiences and configured VoIP systems to enhance communication and productivity.

CERTIFICATES

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|--|---|--|
| • Software Security
(University of Maryland,
College Park) ↗ | • Programming in
Python(Meta) ↗ | • Usable Security
(University of Maryland,
College Park) ↗ |
| • Hardware Security
(University of Maryland,
College Park) ↗ | • Cryptography (University
of Maryland, College
Park) ↗ | • Data, Data,
Everywhere(Google) ↗ |

PROJECTS

Relevant Projects

- **cmscribe:** AI-powered commit message generator that leverages various AI providers to create meaningful commit messages from your code changes (PyPi [↗](#))
- **TiaLinks:** An open-source URL shortener with built-in analytics, helping developers and businesses manage, track, and analyze links easily.
- Designed and built a backend service for an NGO's course management platform, allowing mentors to create courses, assign projects, and support mentee registration and grading, with multiple mentors per course.
- Developed a desktop application leveraging a deep-learning image classifier for accurate facial recognition-based student attendance management, achieving a 98% accuracy rate.
- Worked on a review app for customers to leave feedback on all registered products on the platform

ARTICLES

Email OTP Authentication [↗](#)

10/2024

Medium